Multimedia Signal Processing - 2nd Module

June 26, 2024

1. **Question (8 pts)** Consider two sources X and Y that emit symbols according to the following joint p.m.f.

	x_1	x_2	x_3
y_1	0	1/4	?
y_2	1/16	1/16	1/2

   (a) Estimate the probability of observing the pair (x_3, y_1).

   (b) Define the alphabet of the source X and the alphabet of the source Y.

   (c) Compute the cardinality of the two alphabets.

   (d) Estimate the probability mass functions $p_X(x)$ of X and $p_Y(y)$ of Y.

   (e) Compute the information associated to the symbol x_2.

   (f) Compute the entropy $H(X)$ and the entropy $H(Y)$.

   Consider now the joint source (X, Y) emitting one symbol from X and one symbol from Y at each time instant.

   (g) Compute the joint entropy $H(X, Y)$ using its definition. Comment the achieved result compared to the sum of the marginal entropies $H(X)$ and $H(Y)$.

   (h) Compute the Huffman code, its length, and compare it to the theoretical bound.

2. **Question (3 pts)** Consider the problem of video coding.

   (a) With the help of a block diagram, describe the typical hybrid DPCM/DCT video coding scheme.

   (b) Report a quick definition of each block in terms of input / output.

   (c) Clearly distinguish the encoder from the decoder.

3. **Question (4 pts)** Considering the process of scalar quantization.

   (a) Explain which are the quantities we need for the definition of a scalar quantizer.

   (b) Explain the working principle of Lloyd-Max quantizer, reporting the algorithmic steps.

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Considering a DMS source, it is possible that the Huffman code has the same length of the Shannon code.
   - [T] [F] $H(X,Y) > H(X) + H(Y)$ for independent sources X and Y.
   - [T] [F] Audio tonal masking only happens at some specific frequencies.
   - [T] [F] Consider the problem of synthesizing speech voiced sounds using the source-filter model and a train of pulses as input to the filter. The distance between consecutive pulses affects the pitch of the synthesized voiced sound.
   - [T] [F] The KLT is signal dependent whereas the DCT is not.
   - [T] [F] Consider the quantization of the symbols from a source X. A uniform scalar quantizer is an optimal choice if the symbols of X are uniformly distributed.
   - [T] [F] It is not possible to encode symbols from a DMS source X with rate less than $H(X)$ (i.e., the entropy of the source).
   - [T] [F] Given a fixed rate, there is only one possible achievable distortion according to rate-distortion theory.

5. **Question (Matlab - 10pts)**

   Consider you have an image file called `peppers.png` in your Matlab working directory. This is a RGB image with resolution 512×384.

   Data loading

   (a) Load the image and display it.

   (b) Crop the top-left 256×256 pixels of the image (use the cropped version of the image hereinafter).

   (c) Convert to grayscale and display the image again. Feel free to chose the RGB-to-gray conversion you want.

   Data coding Considering the cropped grayscale image.

   (d) Encode and decode the grayscale image using a uniform scalar quantizer with $\{2, 3, 4, \ldots, 8\}$ bit/symbol.

   (e) For each image version, compute the entropy as a measure of rate.

   (f) For each image version, compute the PSNR as a measure of distortion.

   (g) Plot the rate-distortion curve.